

MTF microdata
(ownership, dayparts)

NASA POWER
solar and temperature

AMDA / SEforALL
cost benchmarks

RURA / EPRA
tariff schedules

Household demand

RAMP appliances +
cooking meals (φ)

Hourly operation

solar, load, battery
(8,760 h $\times$ 20 yr)

Least-cost sizing

unserved demand
 $\leq 5\%$ in every year

Finance

LCOE, NPV and
funding gap

scenarios: share cooking $\times \varphi \times$ tariff $\times$ discount $\times$ random seed